

# Random Forest vs Extra Trees: Titanic Survival Prediction

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## Objective

The objective of this assignment is to understand how tree-based ensemble learning models behave by experimenting with different model parameters. Two ensemble models, Random Forest and Extra Trees, were trained on the Titanic dataset to predict passenger survival.

## Dataset

The Titanic dataset contains passenger information such as passenger class, gender, age, number of siblings, number of parents, and ticket fare. The target variable is *Survived*, which indicates whether the passenger survived or not.

Before training the models, the following preprocessing steps were applied:

- Removed unnecessary columns: Name, Ticket, and Cabin
- Filled missing Age values with the median
- Filled missing Embarked values with the mode
- Converted categorical variable Sex into numeric form
- Applied one-hot encoding to the Embarked feature

## Experiment Setup

Two tree-based ensemble models were tested:

- Random Forest Classifier
- Extra Trees Classifier

The following hyperparameters were tested for both models:

- **Number of trees (`n_estimators`):** 100, 300, 500
- **Maximum tree depth (`max_depth`):** 3, 5

The dataset was split into training and validation sets using an 80/20 split.

## Results

The experiment showed that both models perform well for this classification problem. However, the Extra Trees model achieved slightly higher accuracy in several parameter settings compared to Random Forest.

Extra Trees introduces additional randomness when selecting split points, which can reduce variance and sometimes improve model generalization.

## Discussion

Increasing the number of trees generally improves model stability because predictions are averaged across many trees. However, after a certain point, adding more trees does not significantly increase accuracy and mainly increases computation time.

When the maximum depth of trees is small, the model becomes simpler. This reduces the risk of overfitting but may lead to underfitting because the model cannot capture complex relationships in the data.

## Conclusion

This experiment demonstrates how ensemble learning models behave with different hyperparameter settings. Both Random Forest and Extra Trees are powerful methods for classification problems. In this experiment, the Extra Trees classifier slightly outperformed Random Forest due to its higher randomness during tree construction, which can improve generalization performance.